Oregon TECH

Oregon Institute of Technology

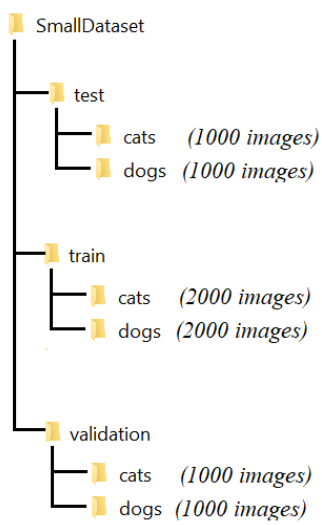
Computer Systems Engineering Technology Department

CST 407 – Python for AI

Homework 2

In this homework you will be using "Dogs vs. Cats" dataset. You can download the original dataset from www.kaggle.com/c/dogs-vs-cats/data (to download the data set you need to create a Kaggle account).

1. After downloading and uncompressing it, create a new dataset called "*SmallDataset*" containing three subsets: a training set with 2,000 unique samples of each class, a validation set with 1000 unique samples of each class, and a test set with 1000 unique samples of each class (all data should come from train data set of the original data set)



2. Verify dimension of your data set by printing out number of images on each directory
3. Preprocess your images:

   -make sure images are converted to 150x150x3

   -make sure image pixels are in float and their magnitudes are between 0 and 1

4. Construct your first NN model: Use five conv layers and one dense layer before the output layer.
5. Choose appropriate loss function, optimizer and metrics.
6. Choose appropriate values for *steps_per_epoch*, *epochs* and *validation_steps.*
7. Run your model
8. Plot training and validation accuracy, training and validation loss
9. Evaluate your model using test data set.
10. Comment on the results.
11. Now add *Dropout()* on your NN model you have created in step 4 and repeat steps 5-9.
12. Use data augmentation to generate more training data from existing training samples
13. Print some samples of the augmented data with the original image.
14. Now run your NN from step 10 by using expanded training data set from step 11
15. Repeat steps 8-10
16. Use pretrained convnet VGG16 (Only use its convolutional-base)
17. Perform the necessary feature extractions using VGG16's convolutional-base
18. Perform the necessary data transformations.
19. Construct your densely connected classifier using single dense layer and a drop out layer before the output layer.
20. Use same the loss function, optimizer and metrics you have chosen in step 5.
21. Run your model.
22. Repeat steps 8-10.
23. Make a general comparison of each method employed.
24. Submit your report and jupyter notebook. Make sure every step is explained in detail.